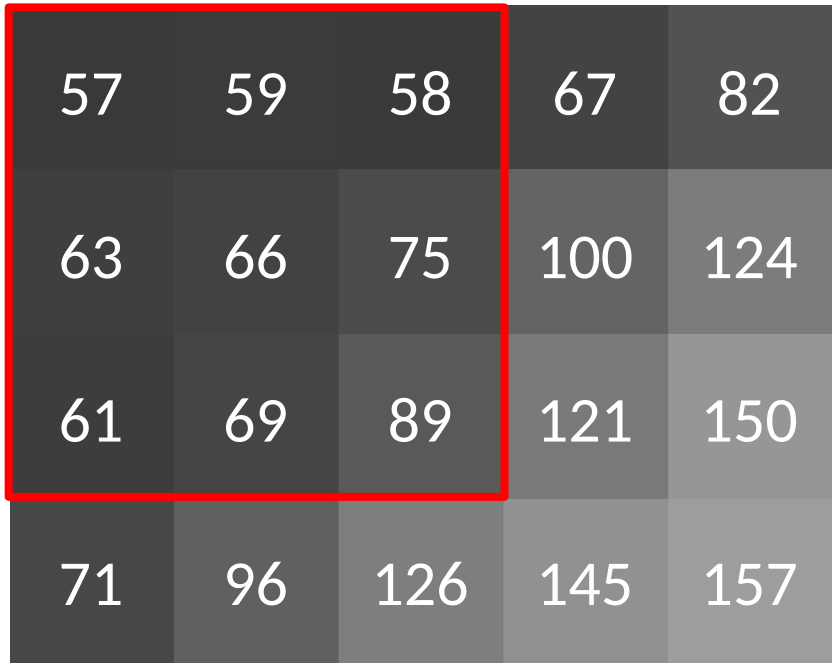
# Computer Vision with Embedded Machine Learning

## Image Convolution and Filtering

© 2021 EdgeImpulse, Inc.

## Image




## Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

(57 · 0) + (59 · -1) + (58 · 0) +
(63 · -1) + (66 · 5) + (75 · -1) +
(61 · 0) + (69 · -1) + (89 · 0) =
64

## Output

| | | |
|---|---|---|
| 64 | | |
| | | |

© 2021 EdgeImpulse, Inc.

Image

| | | | | |
|---|---|---|---|---|
| 57 | 59 | 58 | 67 | 82 |
| 63 | 66 | 75 | 100 | 124 |
| 61 | 69 | 89 | 121 | 150 |
| 71 | 96 | 126 | 145 | 157 |

Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

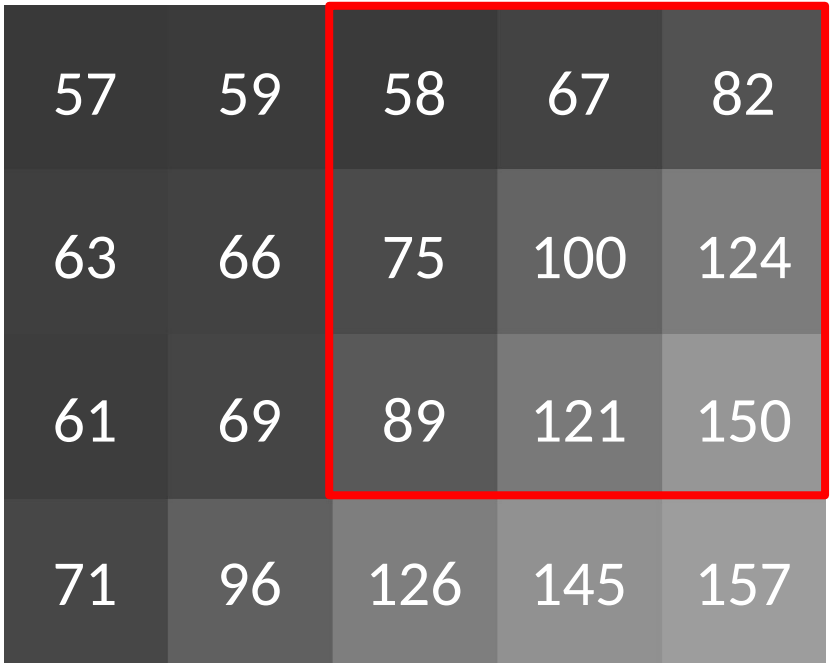
$(59 \cdot 0) + (58 \cdot -1) + (67 \cdot 0) +$
$(66 \cdot -1) + (75 \cdot 5) + (100 \cdot -1) +$
$(69 \cdot 0) + (89 \cdot -1) + (121 \cdot 0) =$
62

Output

| | | |
|---|---|---|
| 64 | 62 | |
| | | |

© 2021 EdgeImpulse, Inc.

Image




Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

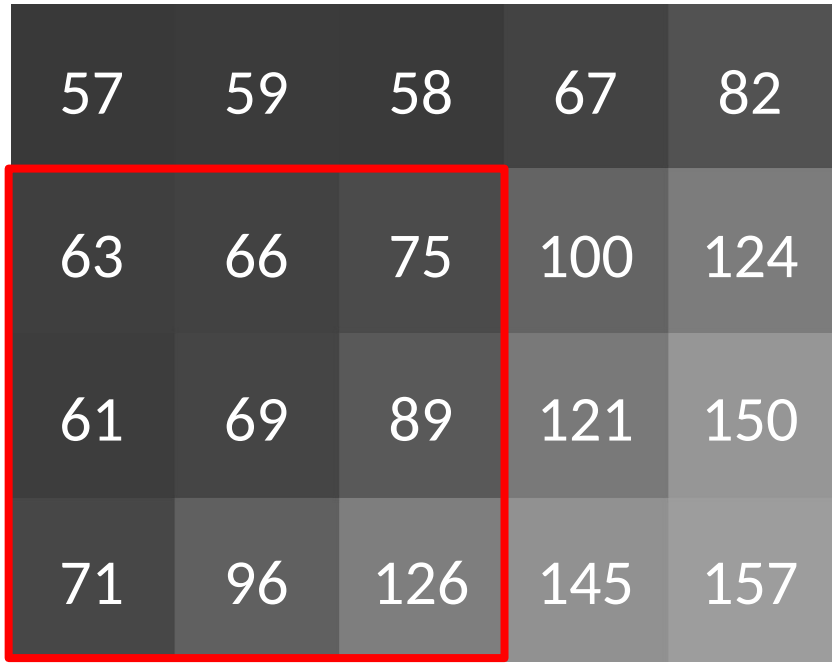
(58 · 0) + (67 · -1) + (82 · 0) +
(75 · -1) + (100 · 5) + (124 · -1) +
(89 · 0) + (121 · -1) + (150 · 0) =
113

Output

| | | |
|---|---|---|
| 64 | 62 | 113 |
| | | |

© 2021 EdgeImpulse, Inc.

Image




Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

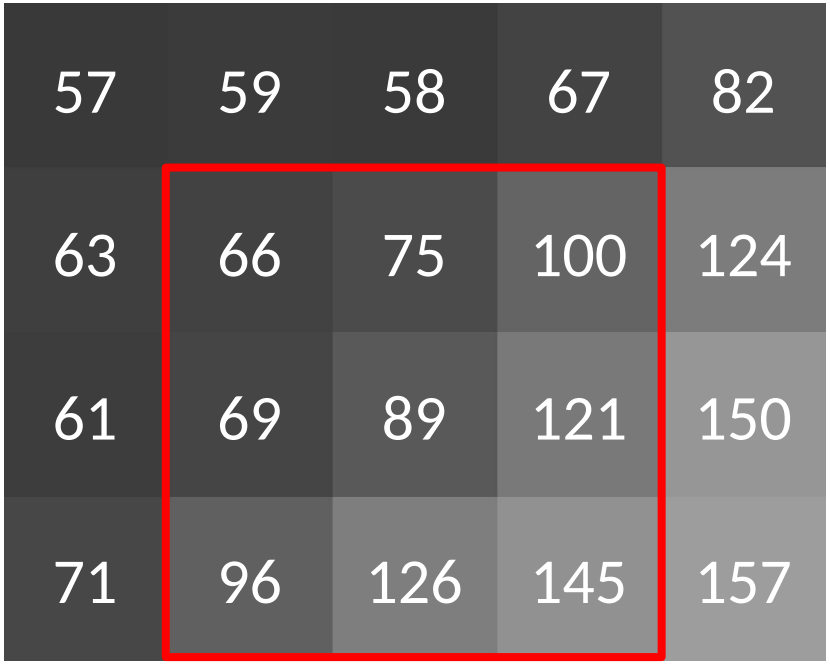
$(63 \cdot 0) + (66 \cdot -1) + (75 \cdot 0) +$
$(61 \cdot -1) + (69 \cdot 5) + (89 \cdot -1) +$
$(71 \cdot 0) + (96 \cdot -1) + (126 \cdot 0) = 33$

Output

| | | |
|---|---|---|
| 64 | 62 | 113 |
| 33 | | |

© 2021 EdgeImpulse, Inc.

Image




Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

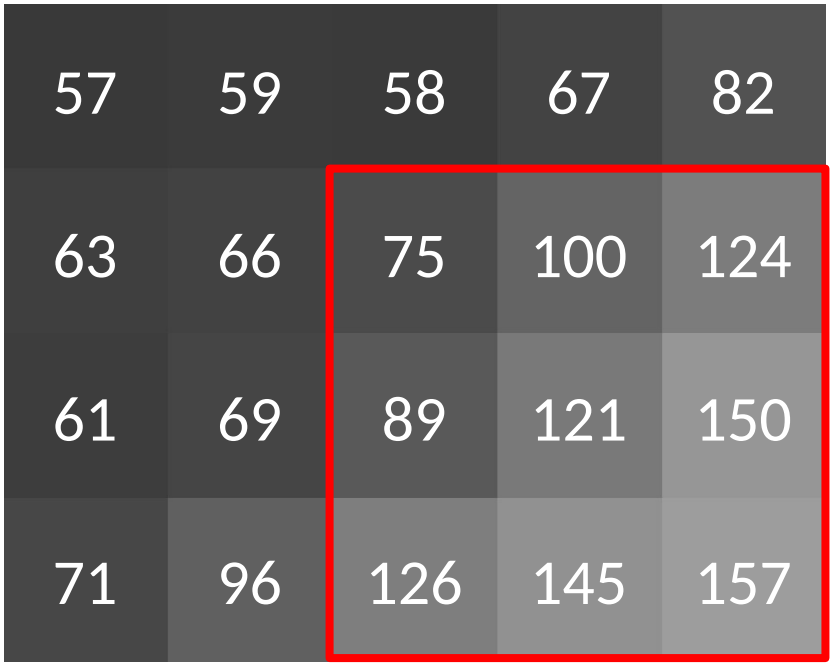
(66 · 0) + (75 · -1) + (100 · 0) +
(69 · -1) + (89 · 5) + (121 · -1) +
(96 · 0) + (126 · -1) + (145 · 0) =
54

Output

| | | |
|---|---|---|
| 64 | 62 | 113 |
| 33 | 54 | |

© 2021 EdgeImpulse, Inc.

Image




stride = 1

Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

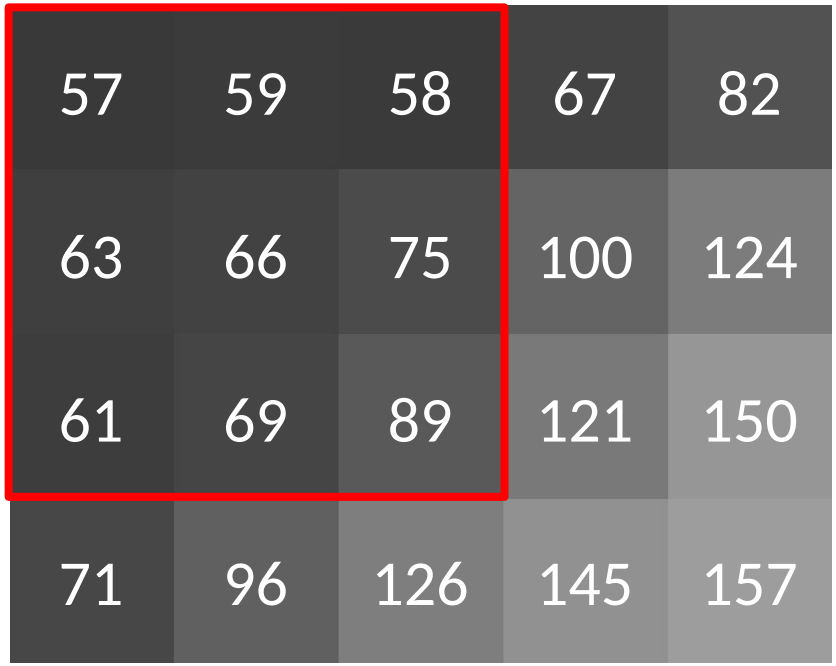
$(75 \cdot 0) + (100 \cdot -1) + (124 \cdot 0) + (89 \cdot -1) + (121 \cdot 5) + (150 \cdot -1) + (126 \cdot 0) + (145 \cdot -1) + (157 \cdot 0) = 121$

Output

| | | |
|---|---|---|
| 64 | 62 | 113 |
| 33 | 54 | 121 |

© 2021 EdgeImpulse, Inc.

Image




stride = 2

Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

$(57 \cdot 0) + (59 \cdot -1) + (58 \cdot 0) +$
$(63 \cdot -1) + (66 \cdot 5) + (75 \cdot -1) +$
$(61 \cdot 0) + (69 \cdot -1) + (89 \cdot 0) =$
64

Output

| | |
|---|---|
| 64 | |

© 2021 EdgeImpulse, Inc.

Image

| | | | | |
|---|---|---|---|---|
| 57 | 59 | 58 | 67 | 82 |
| 63 | 66 | 75 | 100 | 124 |
| 61 | 69 | 89 | 121 | 150 |
| 71 | 96 | 126 | 145 | 157 |

stride = 2

Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

(58 · 0) + (67 · -1) + (82 · 0) +
(75 · -1) + (100 · 5) + (124 · -1) +
(89 · 0) + (121 · -1) + (150 · 0) =
113

Output

| | |
|---|---|
| 64 | 113 |

© 2021 EdgeImpulse, Inc.

$$O(i,j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I((s \cdot i) + m, (s \cdot j) + n) \cdot K(m,n)$$

***s***: Stride

***H***: Number of rows in image (height)
***W***: Number of columns in image (width)

***M***: Number of rows in kernel (height)
***N***: Number of columns in kernel (width)

***i*** goes from 0 to $\left\lfloor \frac{H - M}{s} \right\rfloor + 1$

***j*** goes from 0 to $\left\lfloor \frac{W - N}{s} \right\rfloor + 1$

© 2021 EdgeImpulse, Inc.

$$O(i,j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I((s \cdot i) + m, (s \cdot j) + n) \cdot K(m,n)$$

| | | | | |
|---|---|---|---|---|
| $I_{00}$ | $I_{01}$ | $I_{02}$ | $I_{03}$ | $I_{04}$ |
| $I_{10}$ | $I_{11}$ | $I_{12}$ | $I_{13}$ | $I_{14}$ |
| $I_{20}$ | $I_{21}$ | $I_{22}$ | $I_{23}$ | $I_{24}$ |
| $I_{30}$ | $I_{31}$ | $I_{32}$ | $I_{33}$ | $I_{34}$ |

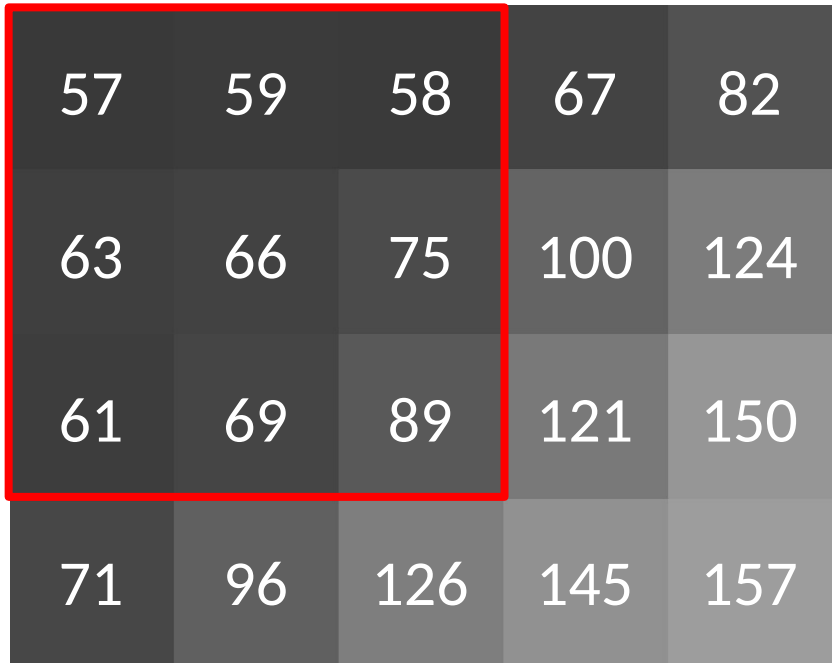
| | | |
|---|---|---|
| $K_{00}$ | $K_{01}$ | $K_{02}$ |
| $K_{10}$ | $K_{11}$ | $K_{12}$ |
| $K_{20}$ | $K_{21}$ | $K_{22}$ |

| | | |
|---|---|---|
| $O_{00}$ | $O_{01}$ | $O_{02}$ |
| $O_{10}$ | $O_{11}$ | $O_{12}$ |

with stride = 1

© 2021 EdgeImpulse, Inc.

Image




8-bit pixel values must be between 0 and 255!

Kernel

| | | |
|---|---|---|
| 1 | 5 | 1 |
| 5 | 5 | 5 |
| 1 | 5 | 1 |

$(57 \cdot 1) + (59 \cdot 5) + (58 \cdot 1) +$
$(63 \cdot 5) + (66 \cdot 5) + (75 \cdot 5) +$
$(61 \cdot 1) + (69 \cdot 5) + (89 \cdot 1) =$
1925

Output

| | | |
|---|---|---|
| 255 | | |
| | | |

© 2021 EdgeImpulse, Inc.

Image

| | | | | |
|---|---|---|---|---|
| 57 | 59 | 58 | 67 | 82 |
| 63 | 66 | 75 | 100 | 124 |
| 61 | 69 | 89 | 121 | 150 |
| 71 | 96 | 126 | 145 | 157 |

8-bit pixel values must be between 0 and 255!

Kernel

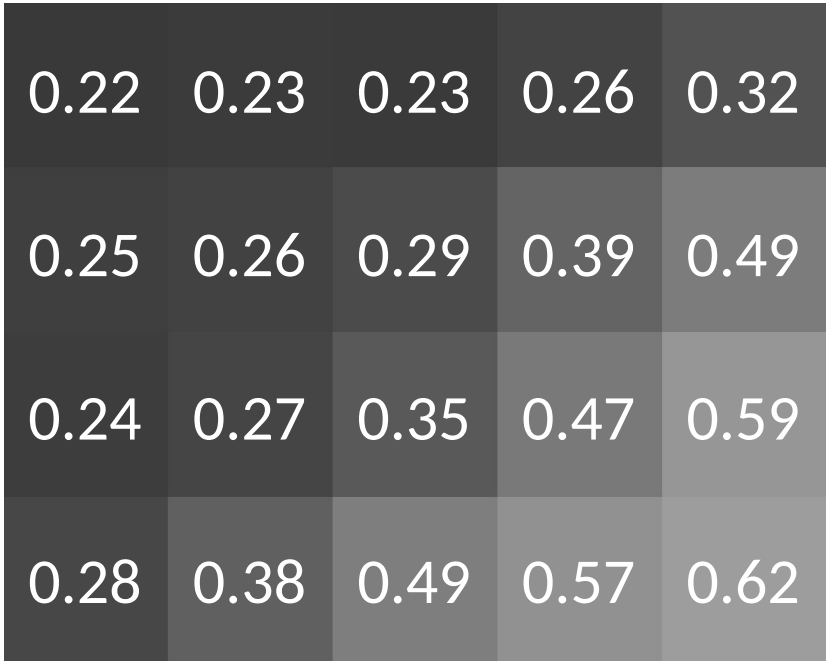
| | | |
|---|---|---|
| 1 | 5 | 1 |
| 5 | 5 | 5 |
| 1 | 5 | 1 |

Output

| | | |
|---|---|---|
| 255 | 255 | 255 |
| 255 | 255 | 255 |

© 2021 EdgeImpulse, Inc.

Image




Floating point pixel values will likely be between 0.0 and 1.0

Kernel

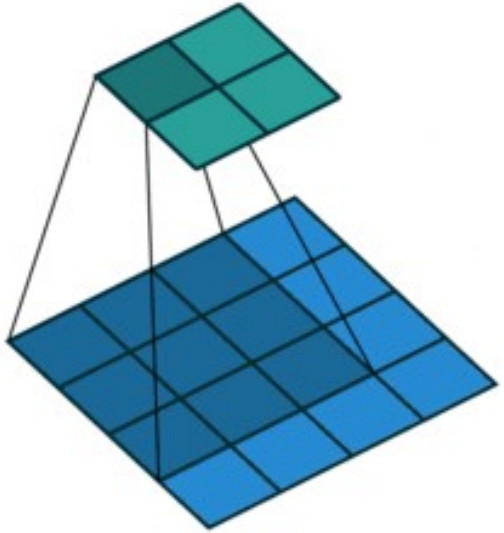
| | | |
|---|---|---|
| 1 | 5 | 1 |
| 5 | 5 | 5 |
| 1 | 5 | 1 |

Output

| | | |
|---|---|---|
| 1.0 | 1.0 | 1.0 |
| 1.0 | 1.0 | 1.0 |

© 2021 EdgeImpulse, Inc.

# Valid Padding

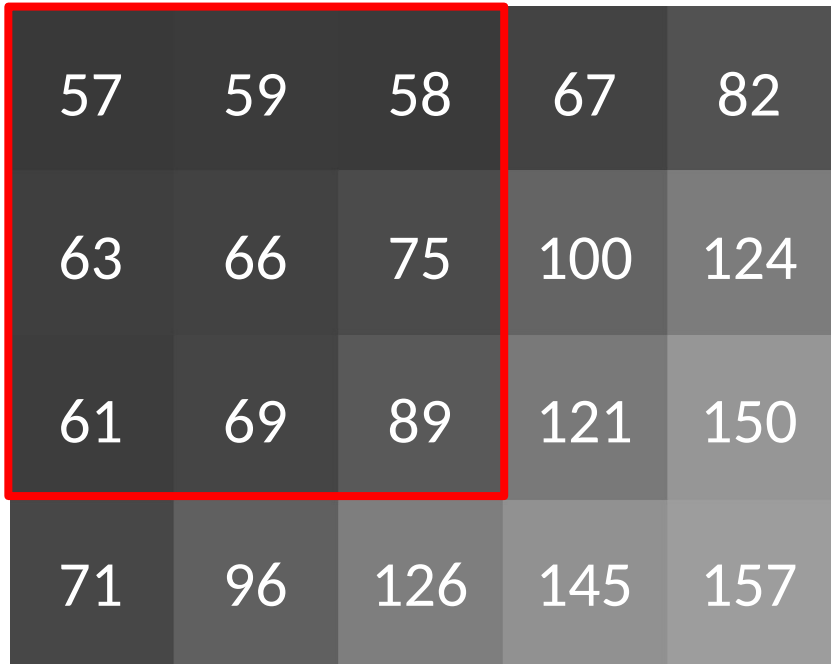


© 2021 EdgeImpulse, Inc.

Animations from github.com/vdumoulin/conv_arithmetic (MIT license)

# Valid Padding

Image




stride = 2

Kernel

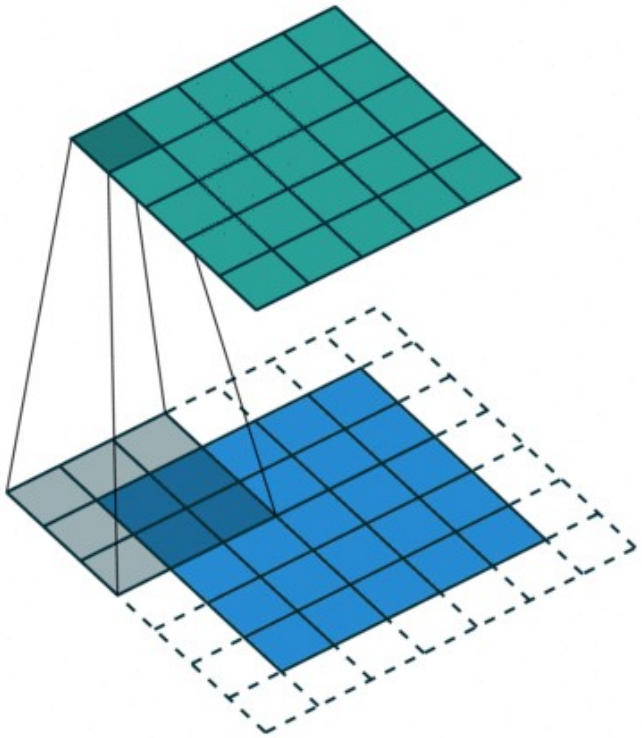
| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

Output

| | |
|---|---|
| 64 | 113 |

© 2021 EdgeImpulse, Inc.

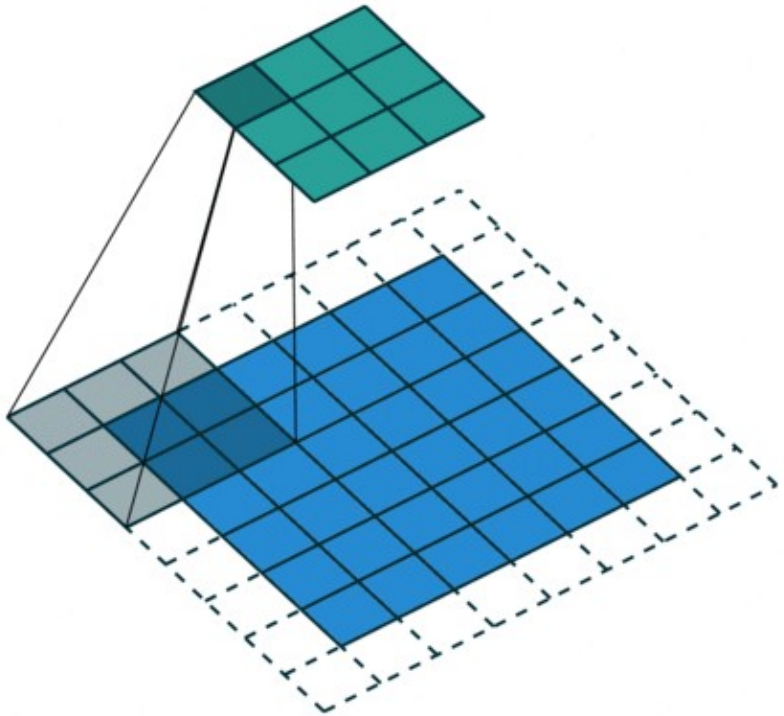
# Same Padding



© 2021 EdgeImpulse, Inc.

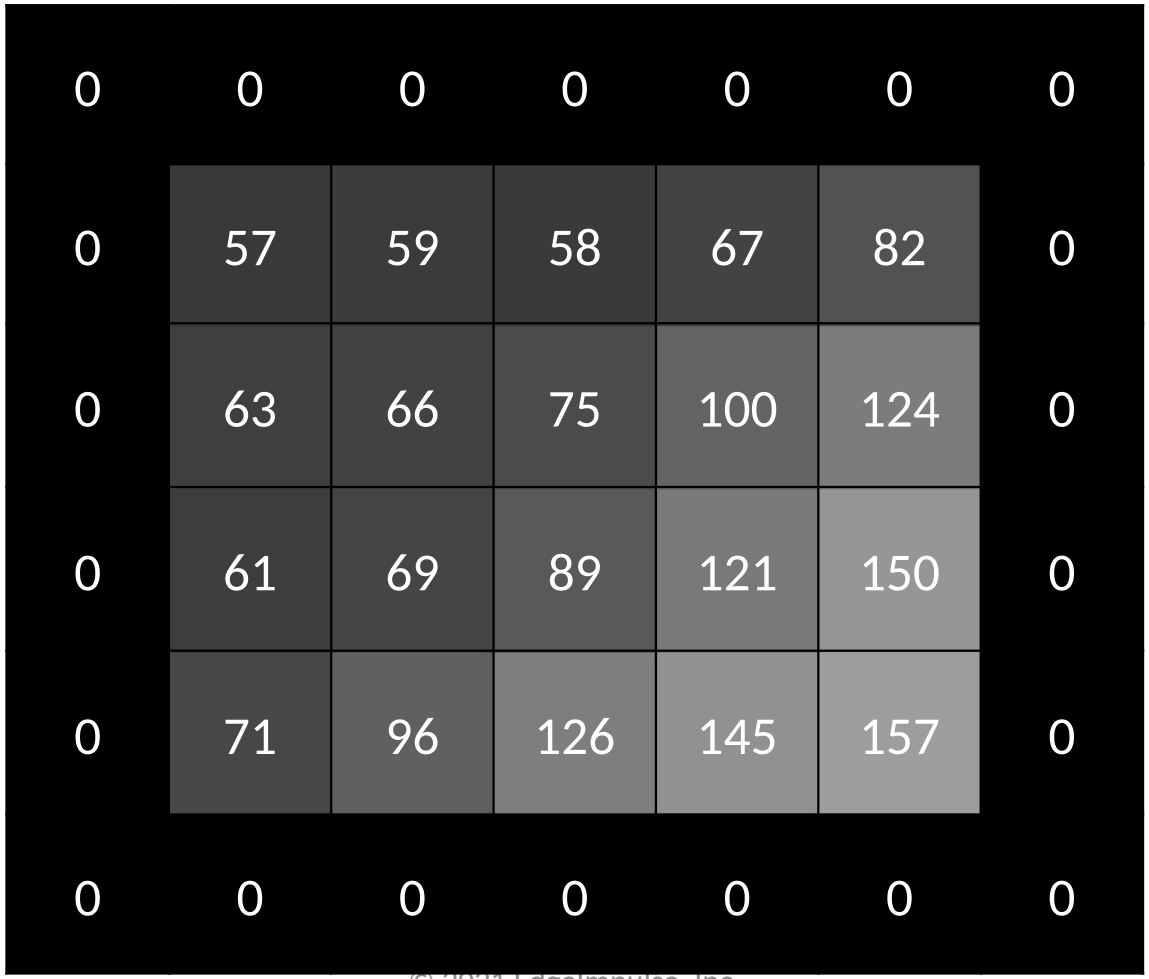
Animations from github.com/vdumoulin/conv_arithmetic (MIT license)

# Padding with stride=2

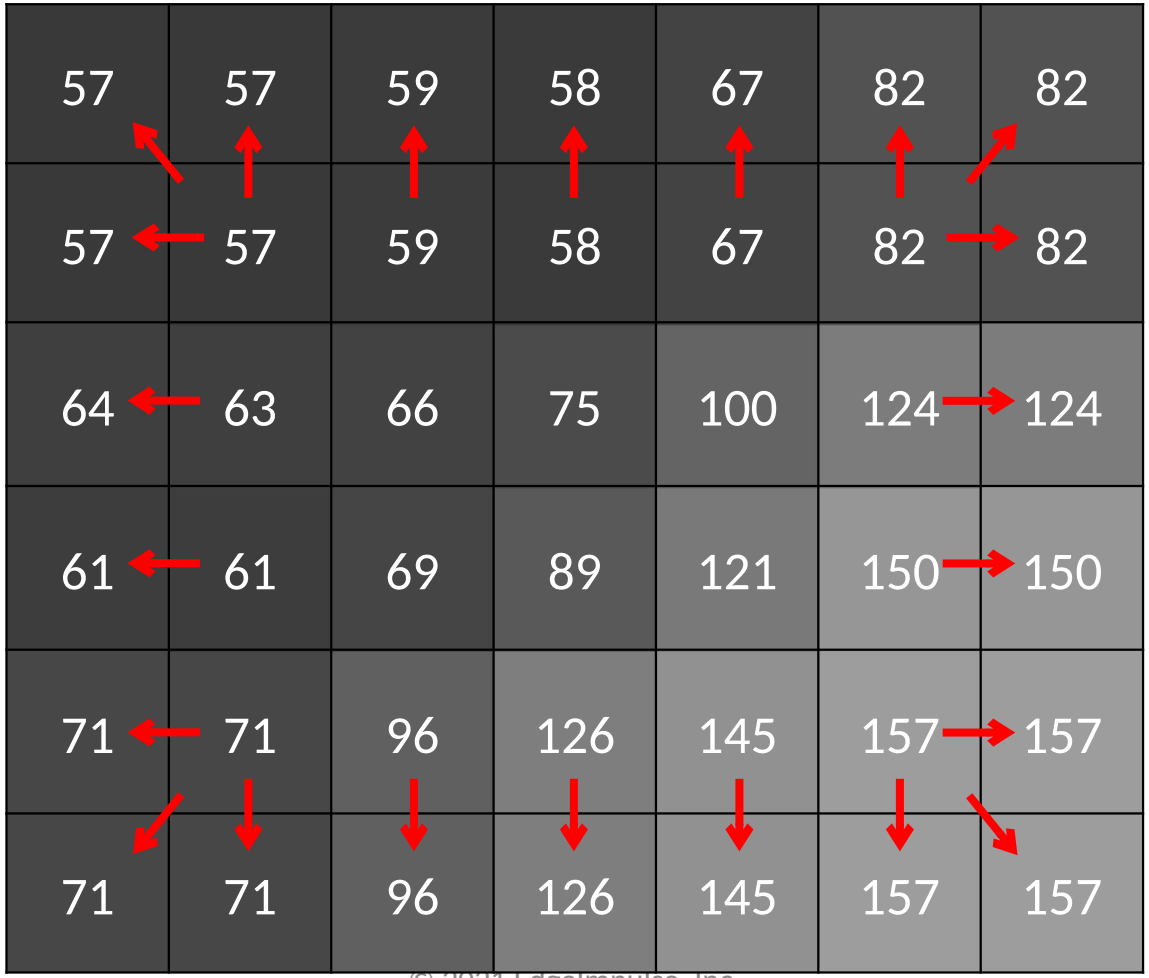


© 2021 EdgeImpulse, Inc.

Animations from github.com/vdumoulin/conv_arithmetic (MIT license)

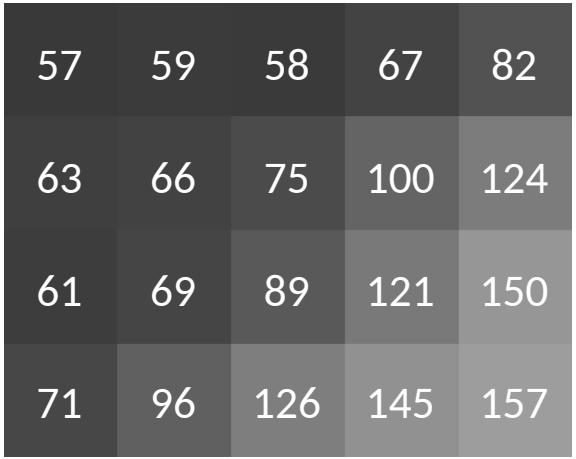



© 2021 EdgeImpulse, Inc.




© 2021 EdgeImpulse, Inc.

## Image (without padding)




stride = 1

## Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

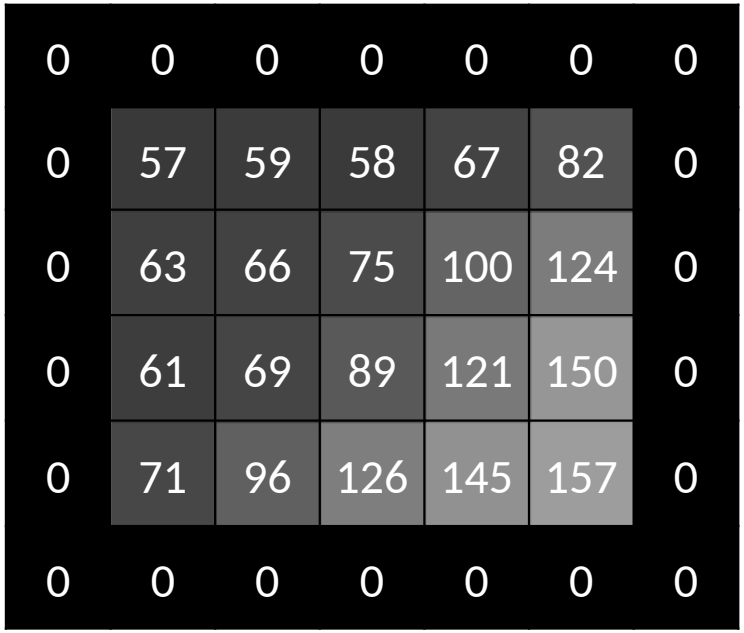
## Output

| | | |
|---|---|---|
| 64 | 62 | 113 |
| 33 | 54 | 121 |

Without padding:
- Smaller output
- Information on borders is lost

© 2021 EdgeImpulse, Inc.

## Input Image (with padding)




stride = 1

## Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

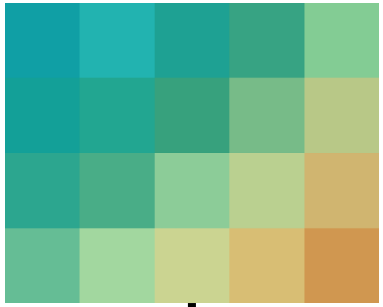
With padding:
- Larger output matrix
- Information on borders is maintained

## Output

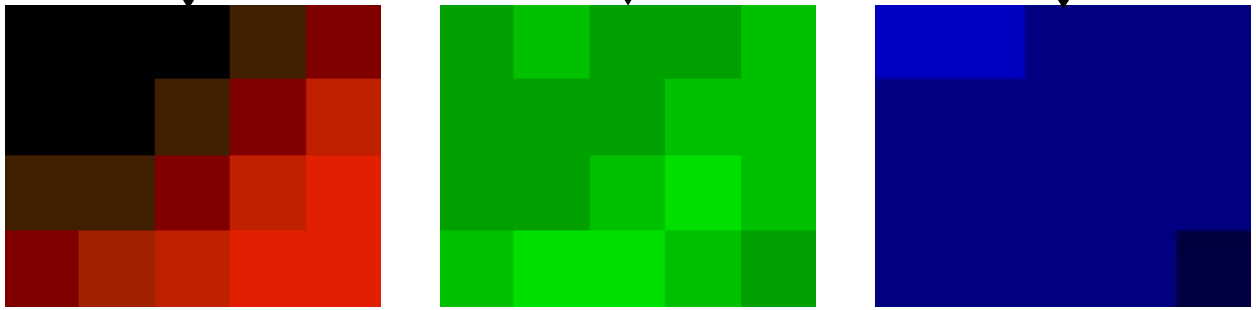
| | | | | |
|---|---|---|---|---|
| 163 | 114 | 89 | 95 | 219 |
| 131 | 64 | 62 | 113 | 255 |
| 102 | 33 | 54 | 121 | 255 |
| 198 | 214 | 255 | 255 | 255 |

© 2021 EdgeImpulse, Inc.

Original image



RGB channels



Kernel

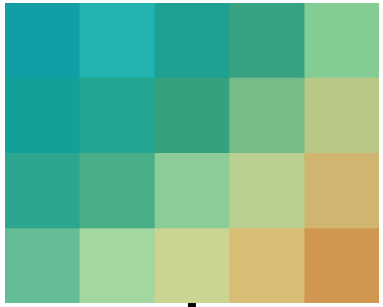
| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

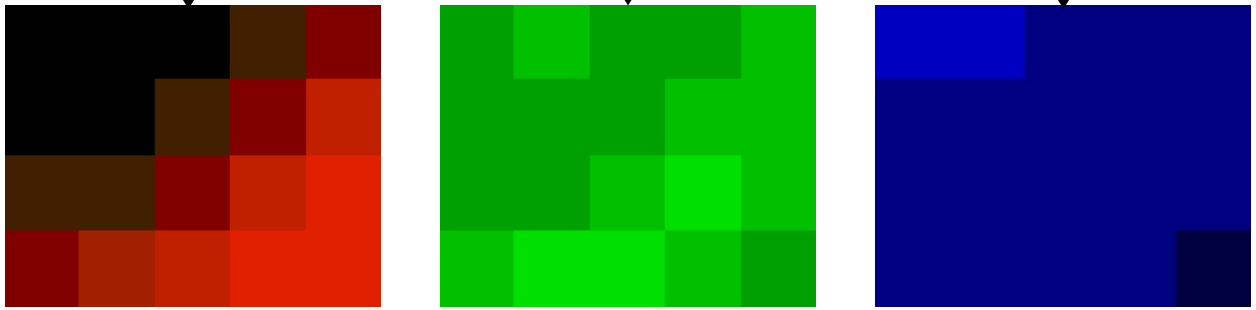
| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

© 2021 EdgeImpulse, Inc.

Original image



RGB channels



Kernel

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |

| | | |
|---|---|---|
| -1 | -2 | -1 |
| 0 | 0 | 0 |
| 1 | 2 | 1 |

| | | |
|---|---|---|
| -1 | -1 | -1 |
| -1 | 8 | -1 |
| -1 | -1 | -1 |

© 2021 EdgeImpulse, Inc.

Original image (200x130)



Gaussian blur

| | | |
|---|---|---|
| 1/16 | 1/8 | 1/16 |
| 1/8 | 1/4 | 1/8 |
| 1/16 | 1/8 | 1/16 |



Outline

| | | |
|---|---|---|
| -1 | -1 | -1 |
| -1 | 8 | -1 |
| -1 | -1 | -1 |



Sharpen

| | | |
|---|---|---|
| 0 | -1 | 0 |
| -1 | 5 | -1 |
| 0 | -1 | 0 |



Left Sobel

| | | |
|---|---|---|
| 1 | 0 | -1 |
| 2 | 0 | -2 |
| 1 | 0 | -1 |



Emboss

| | | |
|---|---|---|
| -2 | -1 | 0 |
| -1 | 1 | 1 |
| 0 | 1 | 2 |



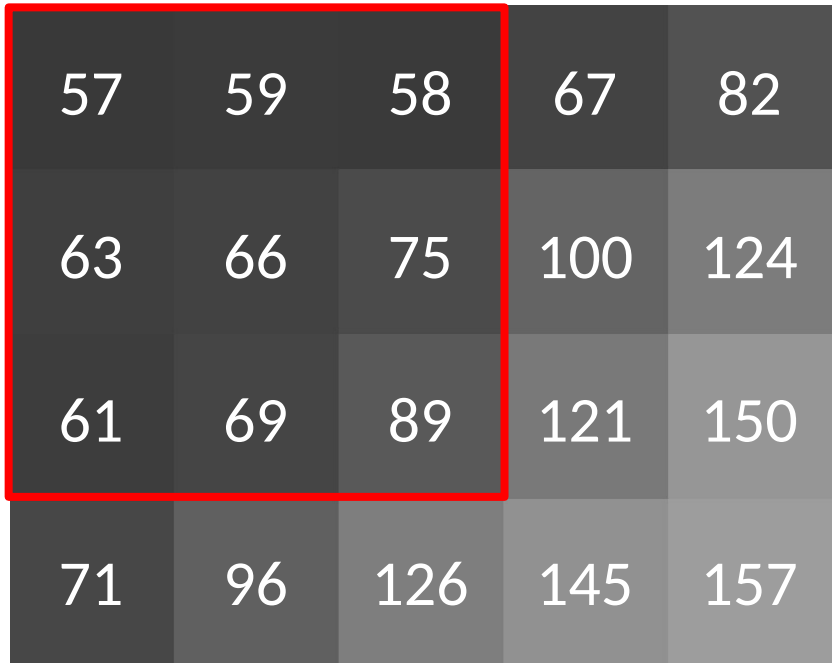
Top Sobel

| | | |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 0 | 0 |
| -1 | -2 | -1 |



© 2021 EdgeImpulse, Inc.

Image




8-bit pixel values must be between 0 and 255!

Kernel

| | | |
|---|---|---|
| 1 | 5 | 1 |
| 5 | 5 | 5 |
| 1 | 5 | 1 |

$(57 \cdot 1) + (59 \cdot 5) + (58 \cdot 1) +$
$(63 \cdot 5) + (66 \cdot 5) + (75 \cdot 5) +$
$(61 \cdot 1) + (69 \cdot 5) + (89 \cdot 1) =$
1925

Output

| | | |
|---|---|---|
| 255 | | |
| | | |

© 2021 EdgeImpulse, Inc.